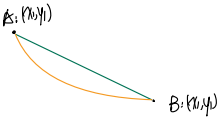
- Unconstrained

Eg.



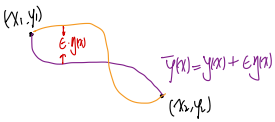
minimize distance

$$\min_y I[y] = \int_{x_0}^{x_1} \sqrt{1+y'(x)^2}\, dx$$

minimize the time from A to B under gravity

$$\tfrac{1}{2}mv^2 = mg(y-y_0)$$

$$\min_y I[y] = \int_{x_0}^{x_1} \frac{\sqrt{1+y'(x)^2}}{v}\, dx = \int_{x_0}^{x_1} \frac{\sqrt{1+y'(x)^2}}{\sqrt{2g(y-y_0)}}\, dx$$



$$\min_y \int_{x_0}^{x_1} F(x, y(x), y'(x))\, dx$$

$$\text{St} \quad y(x_0) = y_0 \quad y(x_1) = y_1$$

let
- $y(x)$ be the optimal solution
- $\bar{y}(x) = y(x) + \varepsilon\eta(x)$
- η be an arbitrary variation satisfying $\eta(x_0) = \eta(x_1) = 0$

should have $\frac{d}{d\varepsilon} \int_{x_0}^{x_1} F(x, \bar{y}(x), \bar{y}'(x))\, dx \Big|_{\varepsilon=0} = 0$

$$\frac{d}{d\varepsilon} \int_{x_0}^{x_1} F(x, \bar{y}(x), \bar{y}'(x))\, dx \Big|_{\varepsilon=0}$$

$$= \int_{x_0}^{x_1} \frac{\partial F}{\partial x} \underbrace{\frac{dx}{d\varepsilon}}_{0} + \frac{\partial F}{\partial y(x)} \eta(x) + \frac{\partial F}{\partial y'(x)} \eta'(x)\, dx$$

$$= \int_{x_0}^{x_1} \frac{\partial F}{\partial y(x)} \eta(x) + \frac{\partial F}{\partial y'(x)} \eta'(x)\, dx \quad \left(\text{Iso Variation weak form since we have } \eta'(x) \text{ in the integral}\right)$$

$$= \int_{x_0}^{x_1} \frac{\partial F}{\partial y(x)} \eta(x)\, dx + \frac{\partial F}{\partial y'(x)} \eta(x) \Big|_{x_0}^{x_1} - \int_{x_0}^{x_1} \eta(x)\, d\frac{\partial F}{\partial y'(x)}$$

$$= \int_{x_0}^{x_1} \eta(x) \left[\frac{\partial F}{\partial y(x)} - \frac{d}{dx} \frac{\partial F}{\partial y'(x)} \right] dx$$

$$= 0$$

Since η is arbitrary, must have $\underbrace{\frac{d}{dx} \frac{\partial F}{\partial y'(x)} - \frac{\partial F}{\partial y(x)}}_{\text{Euler-Lagrange Equation}} = 0 \quad \forall x$

Eg $\min_y I[y] = \int_0^1 (y-y')^2\,dx$

St $y(0)=0 \quad y(1)=1$

$F(x,y(x),y'(x)) = (y(x)-y'(x))^2$

$\frac{d}{dx}\frac{\partial F}{\partial y'(x)} - \frac{\partial F}{\partial y(x)} = \frac{d}{dx}[2y'(x)-2y(x)] - 2(y(x)-y'(x))$

$= 2y''(x) - 2y'(x) - 2y(x) + 2y'(x)$

$= 2y''(x) - 2y(x)$

$= 0 \quad \forall x$

$y''-y=0 \Rightarrow y=e^{rx} \Rightarrow r^2=1 \quad r=\pm 1$

$\Rightarrow y = C_1 e^x + C_2 e^{-x}$

$y(0) = C_1 + C_2 = 0$
$y(1) = C_1 + C_2 \frac{1}{e} = 1$
$\Rightarrow C_1(e-\frac{1}{e}) = 1$

$y = 1\cdot \frac{e^x - e^{-x}}{e-\frac{1}{e}} = 1\cdot\frac{\sinh(x)}{\sinh(1)} \qquad \sinh(x) = \frac{1}{2}(e^x - e^{-x})$

· Constrained

$\min_y \int_{x_0}^{x_1} F(x,y(x),y'(x))\,dx$

St $\int_{x_0}^{x_1} C(x,y(x),y'(x))\,dx = L$

$y(x_0)=y_0 \quad y(x_1)=y_1$

$\mathcal{L}(y,\lambda) = \int_{x_0}^{x_1} F(x,y(x),y'(x))\,dx + \lambda\left[\int_{x_0}^{x_1} C(x,y(x),y'(x))\,dx - L\right]$

let
- $y(x)$ be the optimal solution
- $\bar{y}(x) = y(x) + \varepsilon\eta(x)$
- $\eta(x)$ be an arbitrary variation satisfying $\eta(x_0)=\eta(x_1)=0$

Should have $\begin{cases} \frac{d}{d\varepsilon}\mathcal{L}\big|_{\varepsilon=0} = 0 & \forall \eta \text{ s.t. } \eta(x_0)=\eta(x_1)=0 \\ \int_{x_0}^{x_1} C = L \end{cases}$

$\frac{d}{d\varepsilon}\int \mathcal{L}(y,\lambda) = \int_{x_0}^{x_1} \frac{d}{d\varepsilon} F(x,y(x),y'(x)) + \lambda\frac{d}{d\varepsilon} C(x,y(x),y'(x))\,dx$

$= \int_{x_0}^{x_1} \left[\frac{\partial F}{\partial y(x)} + \lambda\frac{\partial C}{\partial y(x)}\right]\eta(x) + \left[\frac{\partial F}{\partial y'(x)} + \lambda\frac{\partial F}{\partial y'(x)}\right]\eta'(x)\,dx$

$= \int_{x_0}^{x_1} \frac{d(F+\lambda C)}{dy(x)}\eta(x)\,dx + \frac{d(F+\lambda C)}{dy'(x)}\eta(x)\Big|_{x=x_0}^{x=x_1} - \int_{x_0}^{x_1}\eta(x)\,d\frac{d(F+\lambda C)}{dy'(x)}$

$= \int_{x_0}^{x_1} \eta(x)\left[\frac{d(F+\lambda C)}{dy(x)} - \frac{d}{dx}\frac{\partial F(y+\lambda C)}{\partial y'(x)}\right]dx$

$= 0 \quad \forall \eta$

$\frac{d}{dx}\frac{d(F+\lambda C)}{dy'(x)} - \frac{d(F+\lambda C)}{dy(x)} = 0$

Eg. $\min_y \int_0^1 y'(x)^2\,dx$ $\quad F(x,y(x),y'(x)) = y'(x)^2$

s.t. $\int_0^1 y(x)\,dx = 2$ $\quad C(x,y(x),y'(x)) = y(x)$

$y(0)=1 \quad y(1)=1$

$$\frac{d}{dx}\frac{\partial(F+\lambda C)}{\partial y'(x)} - \frac{\partial(F+\lambda C)}{\partial y(x)} = \frac{d}{dx}2y'(x) - \lambda = 0$$

$$\Downarrow$$

$y''(x) = \frac{\lambda}{2}$ $\quad y(x) = \frac{\lambda}{4}x^2 + ax + b$

$y(0)=1$ $\quad y(x) = \frac{\lambda}{4}x^2 + ax + 1$

$$\Downarrow$$

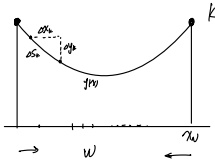
$$\begin{cases} y(1)=1 \Rightarrow \frac{\lambda}{4} + a = 0 \\ \int_0^1 y(x)\,dx = 2 \Rightarrow \left(\frac{\lambda}{12}x^3 + \frac{a}{2}x^2 + x\right)\Big|_0^1 = \frac{\lambda}{12} + \frac{a}{2} + 1 = 2 \end{cases}$$

$$\Downarrow$$

$$\left.\begin{cases} \frac{\lambda}{4} + a = 0 \\ \frac{\lambda}{12} + \frac{a}{2} = 1 \end{cases}\right\} \Rightarrow \lambda = -24 \quad a = 6$$

$y(x) = -6x^2 + 6x + 1$

Eg. Hanging chain



total cable length: $L = \int_{x_1}^{x_2} \sqrt{1+y'(x)^2}\,dx$

total cable mass: $m = \rho L$

when it hangs down: minimum potential energy

$$U = \int_{x_1}^{x_2} \underbrace{\rho\sqrt{1+y'(x)^2}}_{m}\,\underbrace{g}_{g}\,\underbrace{y(x)}_{h}\,dx$$

$\min_y \int_{x_1}^{x_2} \rho\sqrt{1+y'(x)^2}\,g\,y(x)\,dx$ $\quad F(x,y,y') = \rho\sqrt{1+y'^2}\,g\,y$

s.t. $\int_{x_1}^{x_2} \sqrt{1+y'(x)^2}\,dx = L$ $\quad C(x,y,y') = \sqrt{1+y'^2}$

$y(x_1)=y_1 \quad y(x_2)=y_2$

$$\frac{\partial F}{\partial y} + \lambda\frac{\partial C}{\partial y} - \frac{d}{dx}\frac{\partial F}{\partial y'} - \lambda\frac{d}{dx}\frac{\partial C}{\partial y'} = 0$$

$$\rho g\sqrt{1+y'(x)^2} + 0 - \frac{d}{dx}\frac{\rho g y'(x) y(x)}{\sqrt{1+y'(x)^2}} - \lambda\frac{d}{dx}\frac{y'(x)}{\sqrt{1+y'(x)^2}} = 0$$

$$\frac{d}{dx}\left[(\rho g y(x) + \lambda)\frac{y'(x)}{\sqrt{1+y'(x)^2}}\right] = \rho g\sqrt{1+y'(x)^2}$$

· High order

$$\min_x \int L(x, \dot{x}, \ddot{x})\, dt$$

s.t. $x(t_1) = x_1 \quad x(t_2) = x_2$

$\dot{x}(t_1) = \dot{x}_1 \quad \dot{x}(t_2) = \dot{x}_2$

$$\frac{d}{d\delta x} \int L(x, \dot{x}, \ddot{x})\, dt$$

$$= \int_{t_1}^{t_2} \frac{\partial L}{\partial x} \cdot \delta x + \frac{\partial L}{\partial \dot{x}} \delta \dot{x} + \frac{\partial L}{\partial \ddot{x}} \delta \ddot{x}\, dt$$

$$= \int_{t_1}^{t_2} \frac{\partial L}{\partial x} \delta x\, dt + \frac{\partial L}{\partial \dot{x}} \delta x \Big|_{t=t_1}^{t=t_2} - \int \delta x\, d \frac{\partial L}{\partial \dot{x}} + \frac{\partial L}{\partial \ddot{x}} \delta \dot{x} \Big|_{t=t_1}^{t=t_2} - \int \delta \dot{x}\, d \frac{\partial L}{\partial \ddot{x}}$$

$$= \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x\, dt - \int \delta \dot{x} \frac{d}{dt} \frac{\partial L}{\partial \ddot{x}}\, dt$$

$$= \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x\, dt - \left[\frac{d}{dt} \frac{\partial L}{\partial \ddot{x}} \cdot \delta x \Big|_{t=t_1}^{t=t_2} - \int \delta x\, d \frac{d}{dt} \frac{\partial L}{\partial \ddot{x}}\, dt \right]$$

$$= \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} + \frac{d}{dt^2} \frac{\partial L}{\partial \ddot{x}} \right) \delta x\, dt = 0$$

$$\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} + \frac{d}{dt^2} \frac{\partial L}{\partial \ddot{x}} = 0$$